


Bit - Manipulation

Operators

① AND

a	b	<u>a & b</u>
0	0	0
0	1	0
1	0	0
1	1	1



when you & 1 with any number, digits remain the same.

$$\begin{array}{r} 11001010 \checkmark \\ \& 11111111 \\ \hline 11001010 \checkmark \end{array}$$

② OR

a	b	a OR b
0	0	0
0	1	1
1	0	1
1	1	1

③ XOR (^) (if and only if)
exclusive OR

a	b	a ^ b
0	0	0
0	1	1
1	0	1
1	1	0

Observations

$$a \wedge 1 = \bar{a}$$

$$a \wedge 0 = a$$

$$a \wedge a = 0$$

④ Complement (✓)

$$a = 10110$$

$$\bar{a} = 01001$$

Continued ahead

Number Systems

① Decimal $\rightarrow 0, 1, 2, \dots, 9$ Base: (10)
 $(357)_{10}$, $(10)_{10}$

② Binary $\rightarrow 0 \text{ \& } 1$ Base: 2
 $(10)_{10} = (1010)_2$
 $(7)_{10} = (111)_2$

③ Octal $\rightarrow 0, 1, 2, 3, \dots, 7$ Base: 8

Decimal:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

Octal:

0, 1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 20

$$(9)_{10} = (11)_8$$

④

Hexadecimal:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Base: 16

$$(10)_{10} = (A)_{16}$$

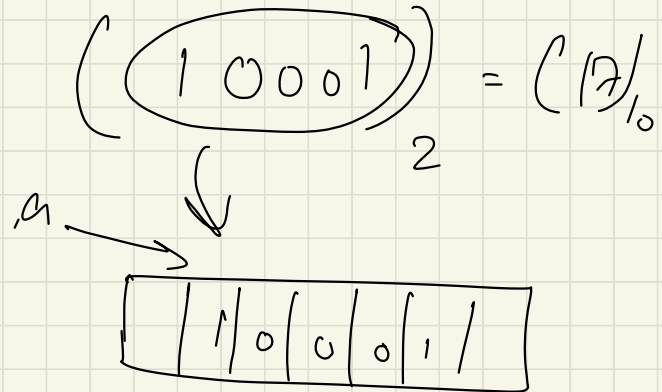
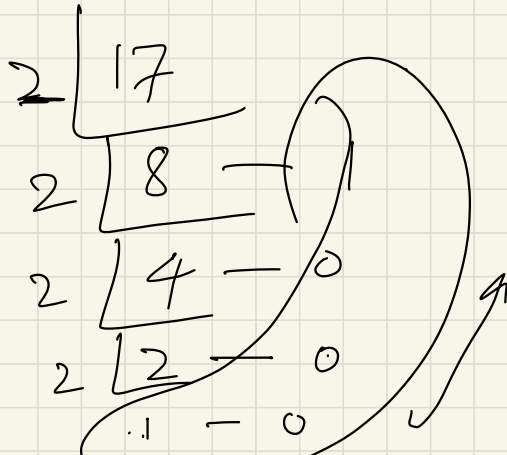
$$(12)_{10} = (C)_{16}$$

Conversions:

① Decimal to base b

Q: Convert $(17)_{10}$ to base 2

Keep dividing by base, take remainders, write in opposite.



$$(17)_{10} = (?)_8 \Rightarrow (21)_8$$

$$\begin{array}{r} 8 \overline{) 17} \\ \underline{16} \\ 1 \end{array}$$

Q Convert any base b to decimal.

$$(10001)_2 = ()_{10} ?$$

Steps:

multiply & add the power of base with digits.

$$= 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 16 + 0 + 0 + 0 + 1 = (17)_{10}$$

$$\underline{\underline{Q:}} \quad (21)_8 = (\quad)_{10}$$

$$= 2 \times 8^1 + 1 \times 8^0$$

$$= 2 \times 8 + 1 = (17)_{10} \quad \checkmark$$

Continuing with operators:

⑤ Left shift operator ($\ll$)

$$(10)_{10} = (1010)_2$$

$$10 \ll 1$$

Step:

$$\begin{array}{cccc} \uparrow & \uparrow & \uparrow & \uparrow \\ 1 & 0 & 1 & 0 \end{array} \ll 1 = 10100$$

$$1 \times 2^4 + 0 \times 2^3 + \cancel{1 \times 2^2} + 0 \times 2^1 + 0 \times 2^0 = \cancel{16} + 4$$
$$= 20$$

$$\therefore a \ll 1 = 2a$$

General point:

$$a \ll b = a \times 2^b$$

⑥ Right shift: $>>$

ex:

$$001100 \downarrow >> 1 \Rightarrow 001100$$

$$(00011234)_{16} = (11234)_{16} \quad \begin{matrix} \updownarrow \\ (1100) \end{matrix}$$

(Ignored) $\rightarrow$ Same for all number systems.

General
point:

$$a \gg b = \frac{a}{2^b}$$

Questions:

(i) Given a no. n find if it is odd or even.

Point:

Every no. is calculated in binary form internally.

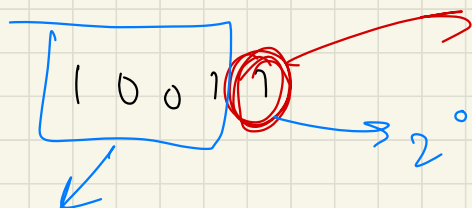
$$12 + 7 \Rightarrow$$

$$(19)_{10} = (10011)_2$$

$$\begin{array}{r} 1100 \\ + 0111 \\ \hline 10011 \end{array}$$

$$16 + 2 + 1 \checkmark$$

Note:



leaving this, every
other is a power of 2

This will always be even.

Notice:

if $2^0 \text{ place} == 1 \Rightarrow$ odd
otherwise $\Rightarrow$ even

$12' + -2''$ + $2'$

Always in even

$1 \rightarrow \text{odd}$
 $0 \rightarrow \text{even}$

$$\begin{array}{r}
 100101 \\
 \& 000001 \\
 \hline
 1001001 \\
 \hline
 n \& 1 == 1 \Rightarrow \text{odd} \\
 \text{else even.}
 \end{array}$$

Sum up:

Q) arr = [2, 3, 4, 1, 2, 1, 3, 6, 4]

Ans

$$(5 \times \cancel{3}) \times (5 \times 4) = 5 \times 5 \times \cancel{3} \times 4$$

$$= 5 \times 4 \times \cancel{3} \times 5$$

$$= 4 \times \cancel{3} \times 5 \times 5$$

$$2 \wedge 3 \wedge 3 \wedge 4 \Rightarrow 2 \wedge 4 \wedge 3 \wedge 5$$

Ans: XOR all the nos.

Q: $[-2, \textcircled{3}, 2, 4, -5, 5, -4]$

Q: Find i^{th} bit of a no.

$\begin{matrix} 8 & 7 & 6 & \textcircled{5} & 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{matrix}$
 ?

This is called a mask.

$n \Rightarrow$ mask with $n-1$ zeros $\rightarrow 1 \ll (n-1)$

$1 \ll 4 \Rightarrow 10000$

Ans: $n \& (1 \ll (n-1))$

Ans:

$\begin{matrix} & & & 5^{\text{th}} & & & & \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{matrix}$

Ans

Q:

Set the i^{th} bit.

→ turn it to 1

$0 \rightarrow 1$
 $1 \rightarrow 1$

4th

	1	0	1	0	1	1	0
or	0	0	0	1	0	0	0

1 0 1 1 1 0

mark

Q:

Reset i^{th} bit:

$1 \rightarrow 0$
 $0 \rightarrow 0$

5th

	1	0	1	0	1	1	0
&	1	1	0	1	1	1	1

1 0 0 0 1 1 0

Ans

How to get this
mask?

mask : $1 \ (1 \leq (n-1))$

Q: Find the position of the right most set bit.

Ex:

1 0 1 1 0 1 0 0
a b
 ↓

Ans = 4

$N = a \mid b$

$-N = \overline{a} \mid b$

$Ans = N \& (-N)$

$a = 101101$

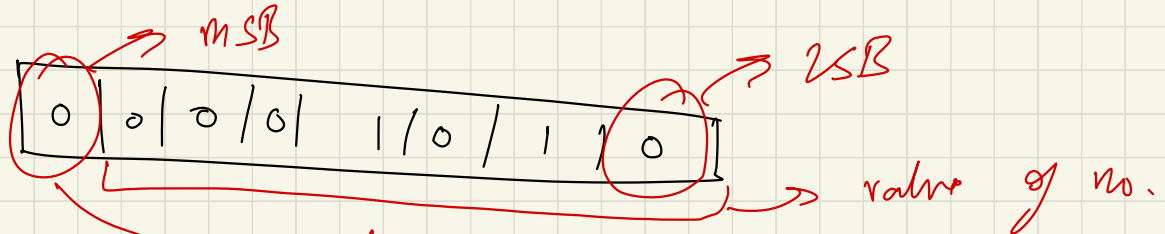
$b = 00$

How?

Negative of a number in Binary form:

1 byte = 8 bits

10 $\Rightarrow$



-10 = ?

$\Rightarrow$ Tells us if no. is +ve or -ve

1 $\rightarrow$ -ve

0 $\rightarrow$ +ve

Steps:

- ① Complement of no.
 - ② + 1 to it
- $\Rightarrow$ aka 2's complement method.

$$(10)_{10} = (00001010)_2$$

$$(1) \quad 11110101$$

$$(2) \quad 11110101$$

+

1

$$\underline{(11110110)_2}$$

$$\Rightarrow (-10)_{10}$$

why two?

10 / 1 0 1 1 0 1 1 1
discarded.

$$\begin{array}{r}
 10000000 \\
 - 00001010 \\
 \hline
 \end{array}$$

What's this?

$$10000000 = 11111111 + 1$$

Now:

$$11111111 + 1 - 00001010$$

$$\Rightarrow 11111111 - 00001010 + 1$$

complement

step 2

$$1000 = 111 + 1$$

$$8 = 7 + 1$$

$$10000 = 1111 + 1$$

$$16 = 15 + 1$$

$$\begin{array}{c}
 111 \\
 1111
 \end{array}$$

$$+ 0001$$

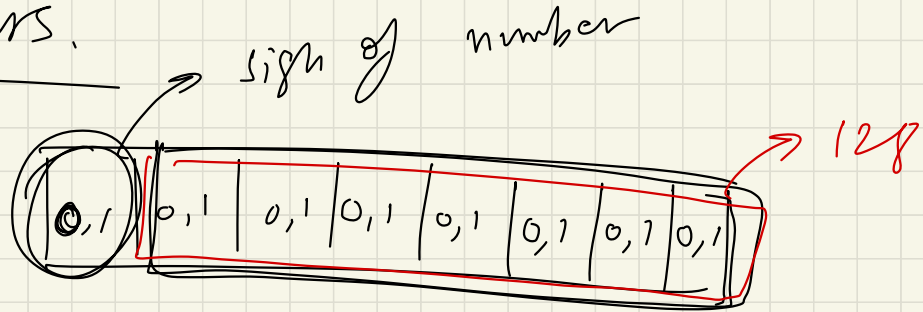
$$\begin{array}{r}
 10000 \\
 \hline
 \end{array}$$

	1	1	1	1	1	1	1	1
—	0	0	0	0	1	0	1	0
	1	1	1	1	0	1	0	1

Complement

Range of numbers.

① 1 byte:



$$\begin{aligned}
 \text{Total} &= 2 \times 2 \times 2 \times \dots \times 2 \text{ times} \\
 &= 2^8 = 256
 \end{aligned}$$

actual no is stored in bits = $n-1$ $\rightarrow$ total bits

1 byte : 7 bits

Total 1 can make from 7 bits
 $= 2^7 = \underline{\underline{128}}$

— 128 to 127

$\rightarrow$ 0 is there as well

0 0 0 0 0 0 0 0

1 1 1 1 1 1 1 1

① $\Rightarrow$

②

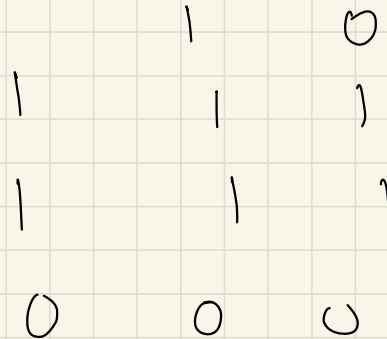
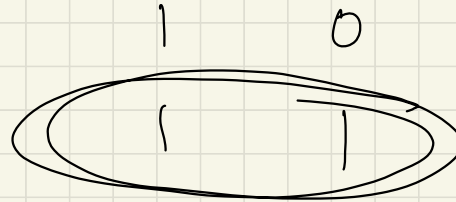
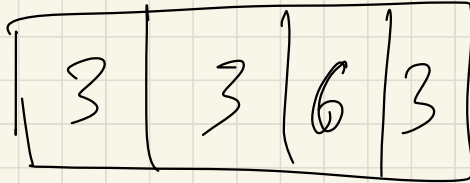
1 1 1 1 1 1 1 1
+ 1

1 0 0 0 0 0 0 0 $\Rightarrow$ 9 bits
will be discarded. $\rightarrow$ -0 = 0

Range formula: for n bits

-2^{n-1} to $2^{n-1} - 1$

Q: arr = [2, 2, 3, 2, 7, 7, 8, 7, 8, 8]



If 3 was not in array.

	1	1	1
1	0	0	0
1	0	0	0

1	1	1	1	1	1	3	3	7	4	% 3
---	---	---	---	---	---	---	---	---	---	-----

0	0	1	1
---	---	---	---

3
16

Q: Find the n^{th} magic no.

$$1 = \begin{matrix} s^3 \\ 0 \end{matrix} \begin{matrix} s^2 \\ 0 \end{matrix} \begin{matrix} s^1 \\ 1 \end{matrix} \longrightarrow \begin{matrix} \text{magic no.} \\ 5 \end{matrix}$$

$$2 = \begin{matrix} 0 \end{matrix} \begin{matrix} 1 \end{matrix} \begin{matrix} 0 \end{matrix} \longrightarrow 2s$$

$$3 = \begin{matrix} 0 \end{matrix} \begin{matrix} 1 \end{matrix} \begin{matrix} 1 \end{matrix} \longrightarrow 3s \quad (s + 2s)$$

$$4 = \begin{matrix} 1 \end{matrix} \begin{matrix} 0 \end{matrix} \begin{matrix} 0 \end{matrix} \longrightarrow 12s$$

$$5 = \begin{matrix} 1 \end{matrix} \begin{matrix} 0 \end{matrix} \begin{matrix} 1 \end{matrix} \longrightarrow 13s$$

$\vdots$

n

$n = 6 \Rightarrow$ 

loop

$n \& 1 \Rightarrow$ This will give me last digit in binary.

$n >> 1$

$$0 \times s^1 + 1 \times s^2 + 1 \times s^3$$

Q: find no. of digits in base b .

$$(6)_{10} \Rightarrow 1$$

$$(6)_{10} = (110)_2 = \textcircled{3}$$

Formula?

$$\log_b a = n$$

$$a = b^n$$

$$\log_2 10 = 3.32$$

$$\log_2 6 = x$$

$$10 = 2^{3.32}$$

$$6 = 2^x$$

int + 1

$\neq$ no. of digits

Formula:

No. of digits in base b of no. n

$$= \text{int}(\log_b n) + 1$$

$$\log_b a = \frac{\log_n a}{\log_n b}$$

Pascal's Triangle :

1				
1	1			
1	2	1		
1	3	3	1	
1	4	6	4	1

1 5 10 10 5 1

∴ Find the sum of n^{th} row.

Ans: Sum of each row =

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

for n^{th} row, sum = 2^{n-1}

Ans:

~~$1 \leq (n-1)$~~ = $1 \times 2^{n-1}$

Ans

Q1

You are given a number. find out if its power of 2 or not.

① 00000, 100010

$$\begin{array}{r} 10000000 = 1111111 + 1 \\ \begin{array}{r} 10000000 \\ \& 0111111 \\ \hline 0 \end{array} \end{array} \quad \begin{array}{r} \begin{array}{c} n-1 \\ \uparrow \\ 1111111 + 1 \\ \hline 10010 \\ 01111 \\ \hline 00100 \end{array} \end{array}$$

Ans: If $n \& (n-1) = 0$ // It is power of 2.

Q: Calculate a^b

$$3^6 \Rightarrow 3 \times 3 \times 3 \times 3 \times 3 \times 3 \quad // O(b)$$

$$3^6 = 3^{110} = 3^{2+4} = 3^2 \times 3^4$$

$$\text{ans} = \cancel{9}$$

$$\text{base} = 3$$

$$\text{base} = 9$$

81

base = base * base

$$O(\log(b))$$

$$n = \cancel{11}0$$

$$n \& 1 \Rightarrow 0$$

$$n \gg 1 \Rightarrow 11 \& 1 \Rightarrow 1$$

$$n = 11 \gg 1 \Rightarrow 11 \& 1 = 1$$

$$n = 1 \gg 1 = 0$$

$$110$$

$$3$$

$$\Rightarrow$$

$$\overset{4}{3} \neq \overset{2}{3} \neq \overset{0}{3} \neq 0$$

$$\text{ans} = \text{ans} * \text{base}$$

Imp to understand this question

Q: Given a number n , find the no. of set bits in it.

$$n = 9$$

$$n = 1001$$

$$\text{Ans} = 2$$

$$n \& (-n) = 0001$$

$$n - [n \& (-n)] = 1000 \Rightarrow 1$$

$$\begin{array}{r}
 n = \quad 1001 \\
 \& \quad 1000 \\
 \hline
 \quad 1000 \quad (1)
 \end{array}$$

$$\begin{array}{r}
 \swarrow \\
 8 \& 7 \Rightarrow \begin{array}{r} 1000 \\ \& 111 \\ \hline 0 \end{array} \quad (2)
 \end{array}$$

No. of set bits = no. of iterations.

Q.:

Find ~~X~~OR of nos from 0 to a.

a	XOR from 0 to a
0	0
1	$0 \wedge 1 = 1$
2	$0 \wedge 1 \wedge 2 = 3$
3	0
4	4
5	1

6	7
7	0
8	8
9	1

If $a \% 4 = 0$

$$a \% 4 = 1$$

$$a \% 4 = 2$$

$$a \% 4 = 3$$

$$\frac{0 \rightarrow a}{a}$$

$$1$$

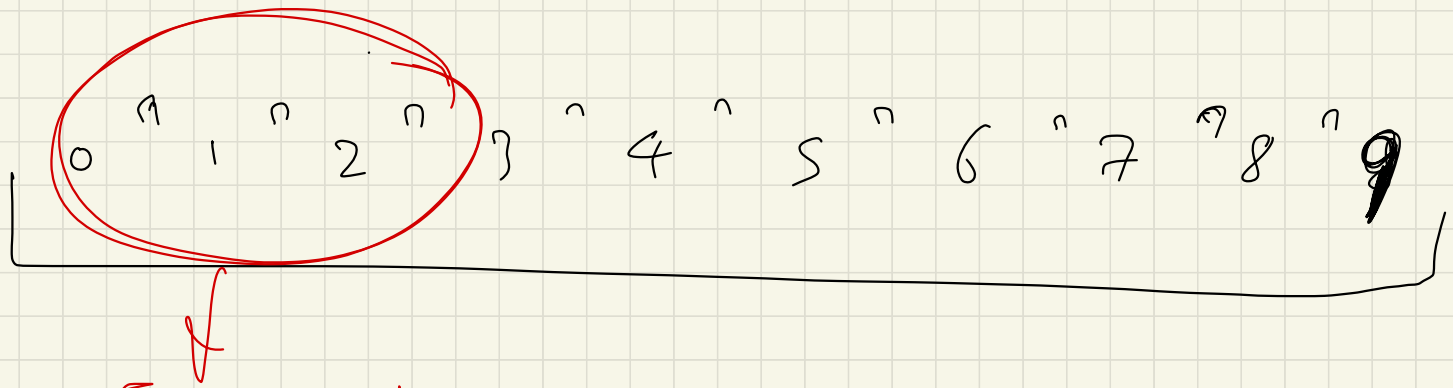
$$a + 1$$

$$0$$

Q: XOR of all nos. between a & b

$$a = 3 \quad \& \quad b = 9$$

$$3 \wedge 4 \wedge 5 \wedge 6 \wedge 7 \wedge 8 \wedge 9$$



These are the extras

This is $0 \rightarrow (a-1)$

Ans =

$$f(b) \wedge f(a-1)$$

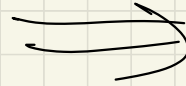
$f(n) \rightarrow$ Xor of $0 \rightarrow x$

Q: Flipping an image:

1 1 0

1 0 1

0 0 0



0 1 1

1 0 1

0 0 0

0 1 1
1 0 1

Flipped

Ans

0 1 1
1 0 1
0 0 0

